

Evaluating Flood Rates Utilizing A GIS Reconstruction of Brooklyn, New York's September 2023 Flood with LiDAR Data

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Abstract:

Situated just 67 meters above sea level, Brooklyn, one of New York City's five boroughs, features subtle elevation fluctuations across its landscape. This topographical diversity, combined with its low elevation and proximity to water bodies, renders Brooklyn highly susceptible to flooding, as evidenced by the events of September 29th, 2023. Given that 2023 marked the hottest year on record and the trajectory of global warming shows no signs of abating, extreme weather occurrences, such as the intense rainstorm experienced in September, are anticipated to become more frequent. It's reasonable to expect another flooding incident in Brooklyn's future. Utilizing GIS software like GlobalMapper and Google Earth, along with high-resolution LiDAR elevation data, enables us to reconstruct and simulate past floods, showing us flood rates and vulnerabilities for different areas, particularly those crucial for emergency services. Armed with this knowledge, residents and community leaders can enhance preparedness for future flood events.

Introduction:

In the last 250 years, carbon emissions from human activities have increased the concentration of CO₂ in the atmosphere by 50%, ushering in an era of rapid global warming, the likes of which have never been seen before in the entire geological history of the planet (EPA, 2022). This global warming leads to extreme weather events and storms that can cause flooding. One such case of flooding occurred in Brooklyn, NY on September 29th, 2023. Photos of the event show mass confusion as entire streets were flooded. Based on the damage done in Brooklyn, it is clear that the region is not prepared for future flooding. Our goals are to help prepare Brooklyn residents for future flooding events by predicting what areas will be most affected by reconstructing the flood in a GIS program so that community leaders can prioritize sewage cleanup and map out what emergency response services will be affected.

Methodology:

- Collected a series of images of the event compiled in an article from the weather channel and determined the geographic location of 9 images (Bonaccorso, 2023)
- Downloaded points as .KMZ files & uploaded them to GlobalMapper
- We drew a polygon around each point to create a “study area” and downloaded LIDAR elevation data (83 files, 66.4 million data points)
- Rendered Bare-Earth DEM “last of many” points to exclude vegetation and buildings to avoid false data
- Simulated water flow in study area using GlobalMapper’s “Flood” tool
- Determined where and how water would accumulate
- All distance and elevation were measured in metric units

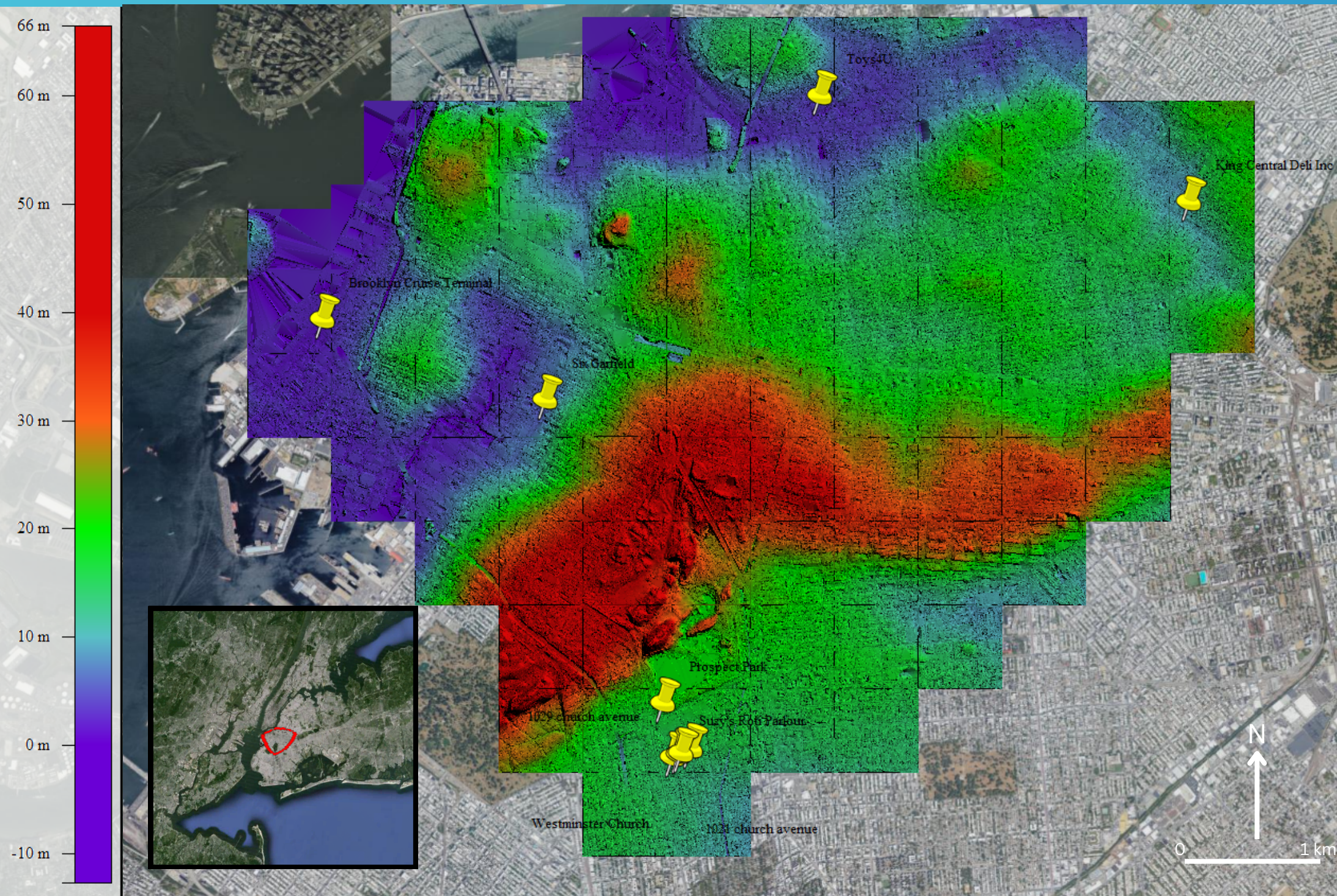


Figure 1: A Global Mapper elevation grid of our study area with 9 labeled points (yellow pins) where floods were reported and LiDAR data displayed. The elevation ranges from 0 meters (blue) to 66 meters (red)

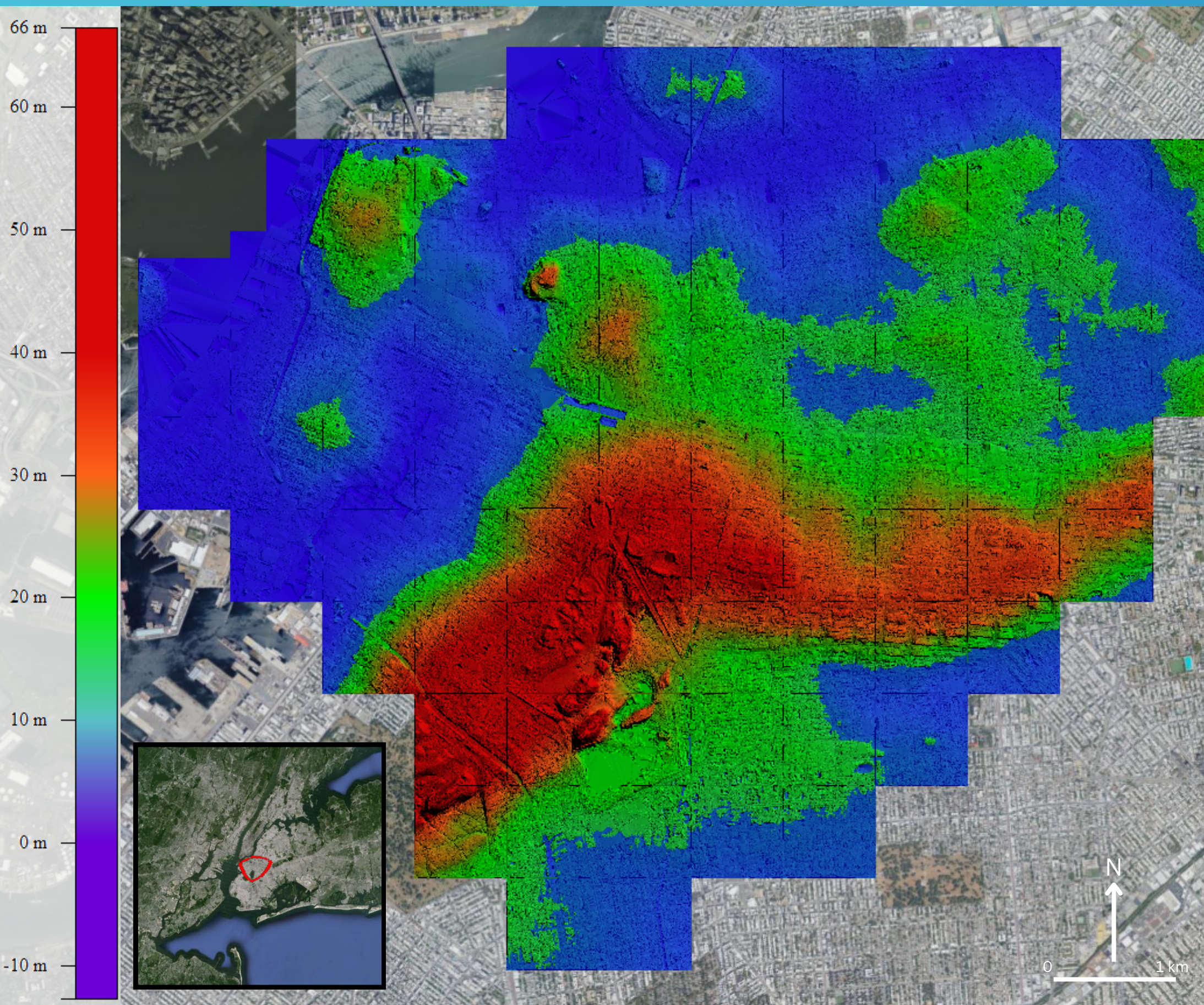


Figure 2: A Global Mapper elevation grid of our study area flood simulated (blue region) with LiDAR data displayed. The elevation ranges from 0 meters (blue) to 66 meters (red)

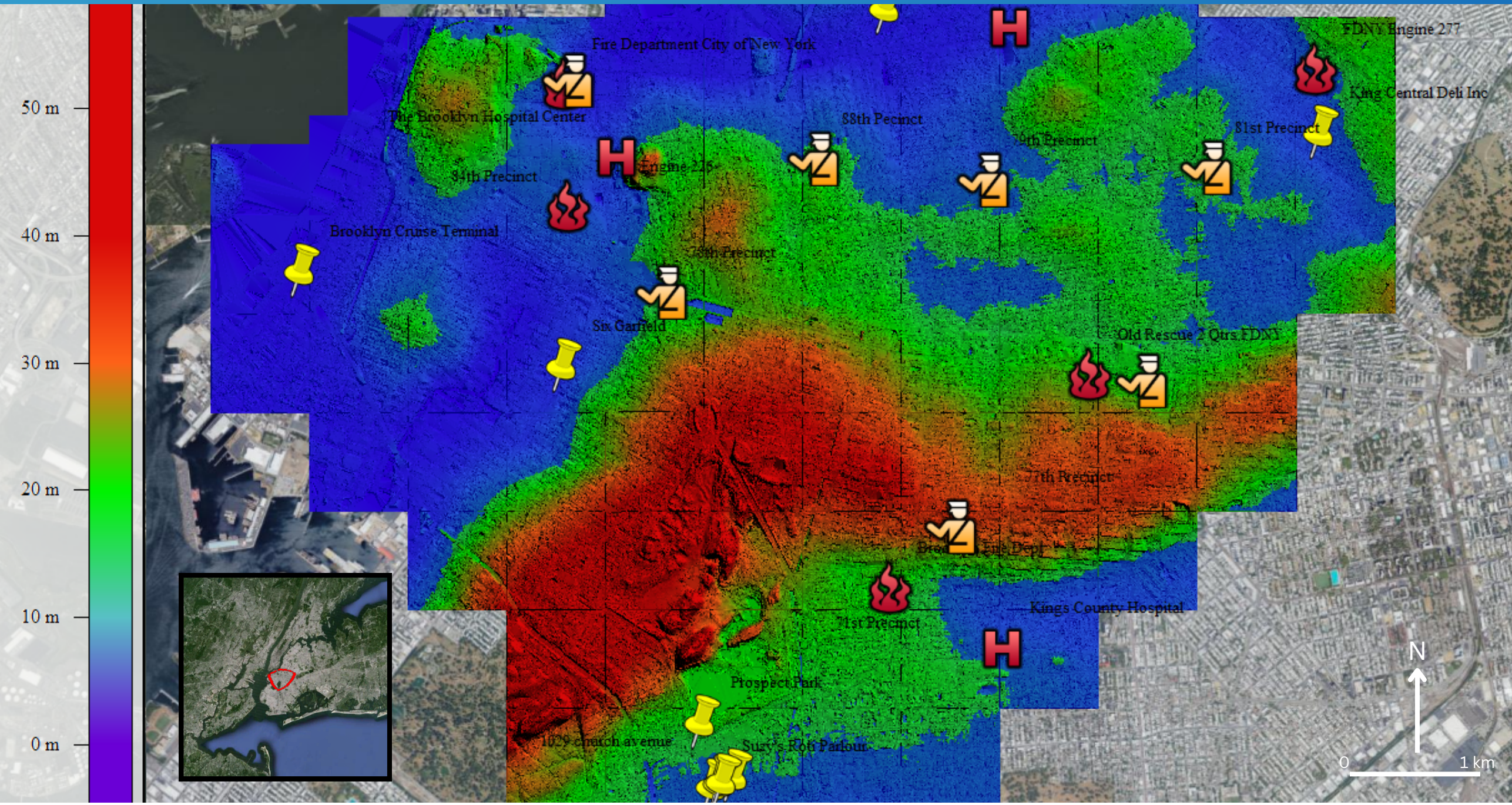


Figure 3: A Global Mapper elevation grid of our study area with flood simulation (blue region), all major rapid response authorities labeled and their vicinity to most affected places, and LiDAR data displayed. The affected roads are also contained within the defined area. The elevation ranges from 0 meters (blue) to 66 meters (red).

Flooded Location	Latitude	Longitude	Elevation (meters)	Estimated Water Depth (meters)
Suzy's Roti Parlour	40°38'47.99"N	73°58'13.53"W	13.1	<0.3
1021 Church Avenue	40°38'48.78"N	73°58'10.91"W	13.1	<0.3
1029 Church Avenue	40°38'49.38"N	73°58'10.02"W	14.9	<0.2
Westminster & Church	40°38'50.46"N	73°58'6.40"W	12.8	<0.1
Prospect Park	40°39'4.07"N	73°58'16.99"W	16.9	<0.2
Six Garfield	40°40'33.25"N	73°59'2.48"W	5.2	<1.0
Brooklyn Cruise Terminal	40°40'57.28"N	74° 0'28.66"W	1.8	<0.6
232 Lee Avenue	40°42'3.03"N	73°57'16.13"W	3	<0.3
King Central Deli Inc	40°41'31.74"N	73°54'53.36"W	9.4	<0.2



Results:

- Many of the roads and intersections were flooded
- Flooding collected at intersections due to dips in elevation
- 10/15 emergency services were flooded
- 4 fire stations
- 3 police precincts
- 3 hospitals
- I-278 was flooded and would have to be closed (evac. Routes)
- Prospect Park peak elevation (66m)
- West->North->South (Long Island shows trends from North to South of a decrease in elevation but there are exceptional localized low-elevation structures such as our study area in Brooklyn)

Discussion/Conclusion:

- Climate change leads to increased flooding; urban areas like Brooklyn must prepare to minimize damage.
- Priority for sewage cleanup should be given to areas of lower elevation.
- Floods can overwhelm sewage systems, contaminating water supply.
- Emergency services need to anticipate increased calls and prepare accordingly.
- Floods impede emergency services, hindering their response to requests for assistance.
- Impending floods may raise insurance costs; precautions can mitigate damage.
- Urban flooding is a global issue, not unique to Brooklyn.
- Coastal settlements worldwide are vulnerable to climate-induced flooding.
- Settlement near coasts historically driven by economic benefits; now poses vulnerability to flooding.
- September 2023 flood affected towns and cities across three states.
- Our data can reasonably extend to the rest of NYC and surrounding areas.
- While global warming may be irrev

References:

- Bonaccorso, Nicole. “Photos of Flooding New York City.” The Weather Channel, 29 Sep. 2023, <https://weather.com/photos/news/2023-09-29-new-york-new-jersey-flood-images>
- “Brooklyn Topographic Map, Elevation, Terrain.” Topographic Maps, [en_gb.topographic-map.com/map-4dsqnh/Brooklyn/?center=40.6446%2C-73.95526&zoom=15&popup=40.65125%2C-73.97075](https://www.topographic-map.com/map-4dsqnh/Brooklyn/?center=40.6446%2C-73.95526&zoom=15&popup=40.65125%2C-73.97075).
- “Lidar.” Gis, gis.ny.gov/lidar.
- Webmaster, CPC. “NOAA’s Climate Prediction Center.” Climate Prediction Center, 1 Jan. 2001, www.cpc.ncep.noaa.gov/.

Credit authorship Contribution Statement:

Lawlor, J.: GIS processing, editing, writing - Abstract, Introduction, Methodology, Discussion, Conclusion, References; Jones, W.: Figures, GIS Processing, editing, writing - Abstract, Introduction, Methodology, Results, References; Tecusan, K.: GIS processing; Marsellos, A.E.: Supervision, GIS processing, guidance, editing